

ECE 394 Report 2

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1 Lab 3 - Inductor Design

1.1 Choice of Parameters

1.1.1 Ripple Ratio and Frequency

The goal of our design is to have a balance between high efficiency and inductor size. More emphasis is placed on efficiency. For our selection procedure, we used the optimization code from the in-class homework assignments. This code produced a sorted table of ripple ratios and frequencies for inductor sizing and efficiency. For our board, we went to the most efficient combinations. These also tended to have the worst inductor sizing. Then we went down five rows and selected the fifth-most-efficient combination. The reason for this is that the first five combinations had relatively similar efficiencies, all around 0.168-0.2% loss, but the next step down had a significant increase to 0.27% loss. Thus, we decided this was the best compromise for efficiency whilst not having the worst inductor sizing.

This selection resulted in a ripple ratio of 0.8 and a frequency of 60 kHz

1.1.2 MOSFET Selection

The MOSFET selection was relatively straightforward. Based on the optimization table from the previous section, we needed an optimal resistance of $0.2\ \Omega$. Looking through the MOSFET options, this clearly led us to the IRFB7534 MOSFET.

1.1.3 Inductor Design

The inductor design was driven using MATLAB code. We developed a short script that enabled us to test different combinations of cores and wire sizes rapidly.

Our design process was as follows:

1. Based on the equations discussed in class, a ripple ratio of 0.8, frequency of 60 kHz, and a total power of 250 W (leading to 10.416 A of current), leads to an inductance of $12\ \mu\text{H}$.
2. Set the efficiency target to be 1% loss.
3. Based on the 1% loss and 10.416 A average current, we can calculate our maximum winding resistance of $7.6\ \text{m}\Omega$.
4. Using the "four inductor equations" discussed in class, we can calculate the needed number of turns, winding resistance, size factor, and air gap needed for each core and wire size combination.

We wanted to minimize our winding resistance (and thereby losses), so we selected the largest wire gauge possible (14 AWG). Then, we tested the different cores.

In this case, we selected the ETD 29 core because it met all the other specs (enough area for the windings, below the maximum winding resistance, above the minimum K_g , etc.) for the inductor while being smaller. This was important to us since while other cores may have

had more headroom on the different parameters, ETD 29 was the smallest in geometry and would look the cleanest on the board (one of our goals)!

Table 1: Final Design Parameters	
Parameter	Value
Inductance	12 μ H
Core	ETD 29
Wire Size	14 AWG
Number of Turns	9 turns
Air Gap	0.68 mm (0.34 mm reality)
K_g	0.1061
Winding Resistance	7.6 m Ω
MOSFET	IRFB7534
Ripple Ratio	0.8
Frequency	60 kHz
Measured Inductance	12.5 μ H
Measured Winding Resistance	11.50 m Ω

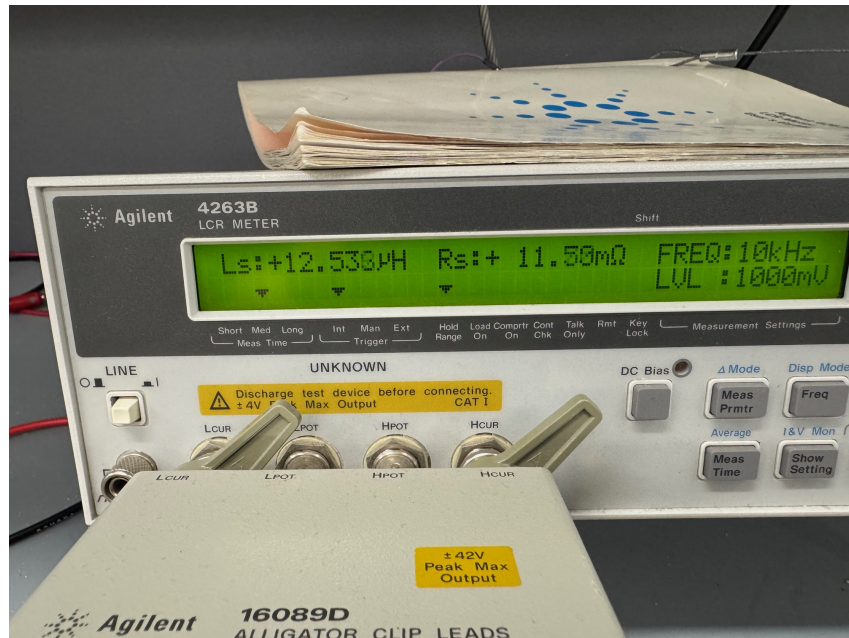


Figure 1: Measured Inductance/Resistance using LCR Meter

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1.2 Difference Between Measured and Theoretical Inductance

Note that there is a small gap between the measured and theoretical inductance. This could be due to several factors. First of all, the air gap is not precisely as calculated, since we did not have the precise shim sizes to achieve it. Furthermore, this does not even account for tolerances in the core's parameters, wire parameters, shim size, etc. The winding may also not be perfectly tight or even. In our calculations, we also largely ignored leakage flux and fringing near the air gap.

The winding resistance is also much higher than the calculated value, as expected. To facilitate soldering and current measurement, the wires extending from the inductor are significantly longer than the winding on the core. We measured from the ends of this wire, which significantly increased the resistance. This can also affect the inductance, as the wire itself has parasitic inductance from the loop it forms with the LCR, along with any self-inductance.

2 Lab 4 - Testing

2.1 Gate Driver Testing

2.1.1 Testing Process and Issues

Before we sent power to our boost converter, we cold-commissioned our board to ensure that all required components were correctly soldered to our PCB: SMD Decoupling Cap, 12V Power Supply, Gate Driver Sub-Circuit, Gate Driver shutdown switch circuit, Pin Headers for the C2000 Launchpad, Input/Output Power Connectors, MOSFETs. Then, we used a DMM (Digital Multimeter) to perform a continuity check on all the pins of these components to ensure that they are not shorted internally or to ground. Next, we reprogrammed the PWM generator on the C2000 Launchpad by adjusting the PWM registers to achieve the switching frequency that we designed for our inductor.

After all the cold-commissioning checklists have been verified, we carefully set the power supply's current limit to prevent the current from jumping to infinity if an internal short occurs on our PCB. Then, we turned on the Power Supply and used the DMM to measure the voltage at 3.3 V pins, $V_{in,sense}$ pins, and $V_{out,sense}$ pins, and prevented these pins' voltage from exceeding 3.3 V, which would damage our C2000 Launchpad. After verifying these pin voltages, we plugged our C2000 Launchpad into the Pin Headers and flashed our updated PWM Generator program to start the Gate Driver Testing Procedure.

At first, we used the Passive Probe connected to the Oscilloscope to measure the ePWM signals V_{PWNA} and V_{PWNB} generated from the C2000 pins.

After verifying that both ePWM signals from the C2000 Launchpad are good, we used differential probes to measure the ePWM signals at the Gate Driver pins and confirm they match those from the C2000 Launchpad.

Next, we closed the Gate Driver Shutdown switch to turn on the Gate Driver. After the Gate Driver is turned on, we used the Differential Probes to measure the bootstrap voltage

waveform by placing one probe at the V_{boot} testing node and the other at the V_{switch} testing node.

After verifying that the bootstrap voltage waveform is good, we used the Passive Probe again to measure the switch node voltage waveform by placing the probe at the V_{switch} test node, referenced to ground.

After verifying that the switch node voltage waveform is good, we used the Passive Probe to measure $V_{gs,low}$ and the Differential Probe to measure $V_{gs,high}$ at the gate driver output. More specifically, to measure the $V_{gs,low}$, we placed the Passive Probe at the Gate Voltage test node of the low-side MOSFETs, referenced to ground. To measure the $V_{gs,high}$ using the Differential Probe, we placed one probe at the Gate Voltage test node of the high-side MOSFETs while placing the other probe at the V_{switch} test node.

We then placed the $V_{gs,low}$ waveform and $V_{gs,high}$ waveform on top of each other on the oscilloscope to measure the Gate Driver dead time, which is the period of time between after $V_{gs,low}$ decreasing to 0 Volt and before $V_{gs,high}$ increasing to 12 Volts and vice versa.

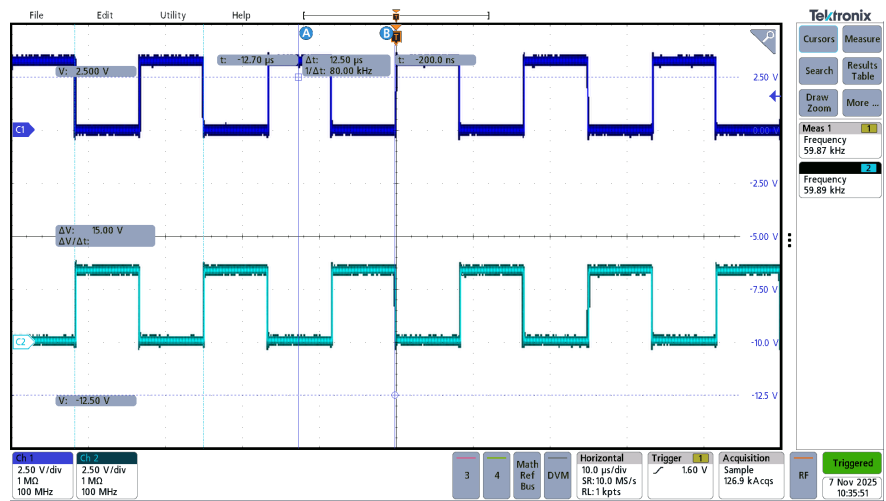


Figure 2: PWM signals from C2000: V_{PWNA} and V_{PWNB}

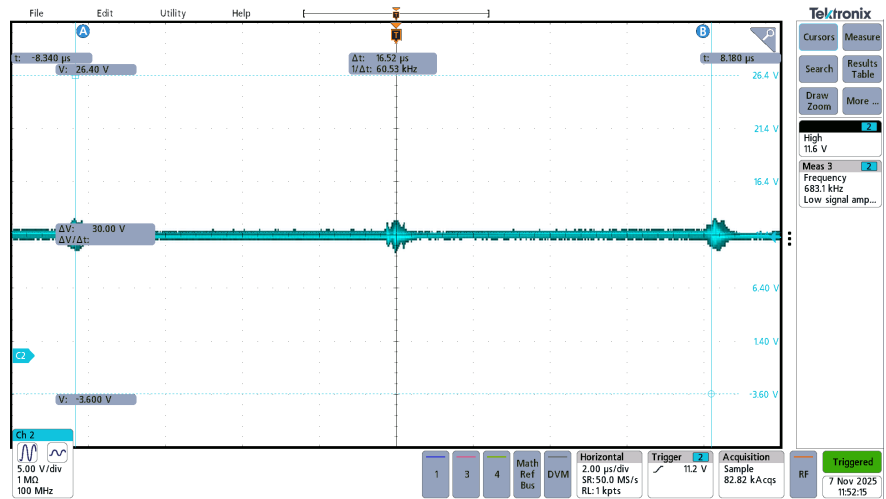


Figure 3: Bootstrap voltage waveform

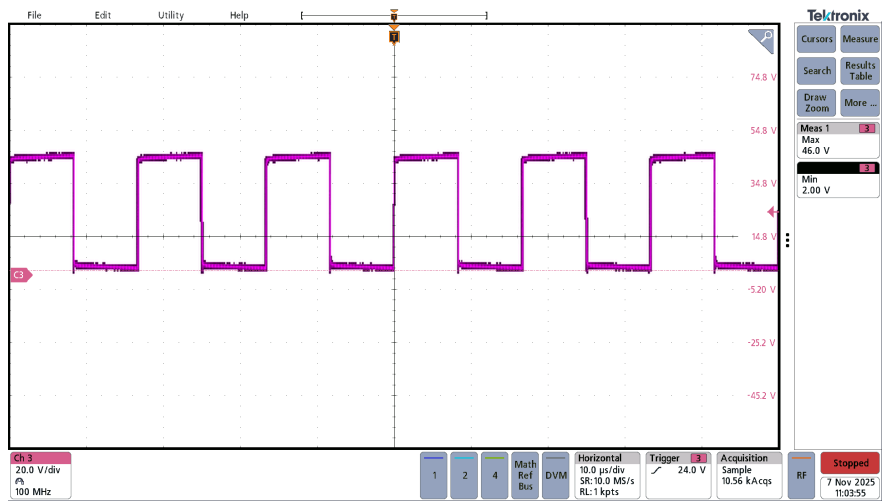


Figure 4: Switch node voltage waveform

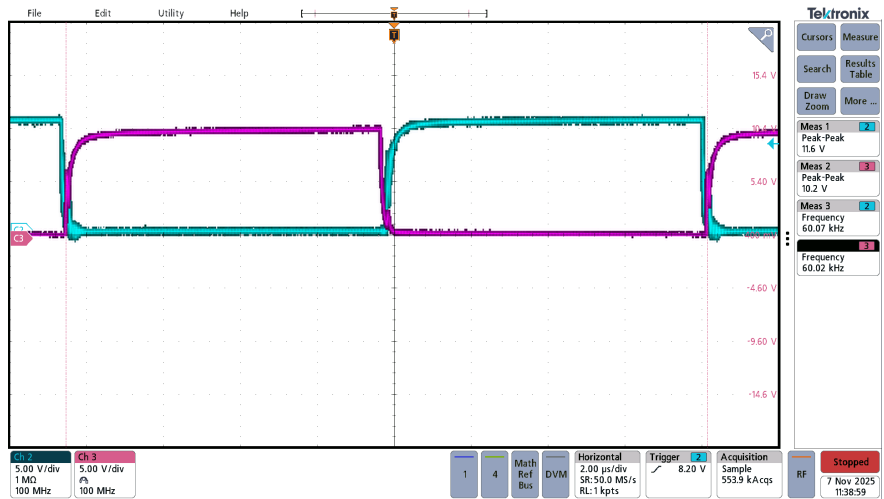


Figure 5: Gate driver output $V_{gs,low}$ and $V_{gs,high}$

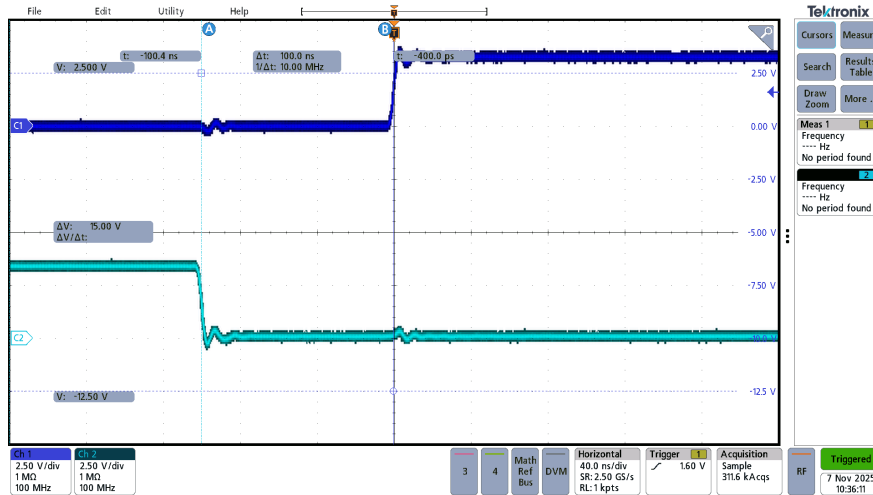


Figure 6: Gate driver dead time

2.2 Boost Under Load Testing

2.2.1 Testing Process and Issues

The testing process for the boost under load was lengthy. The interesting part is that, at first, it worked without issue. We finished soldering all the parts, then connected a low load and turned on the power supply. We used a multimeter at the output to get an easy measurement of the boosted voltage. It returned a perfect output on the first try. However, we quickly had issues regarding the power supplies. We initially slowly ramped up the voltage to 24 V, and that ended up burning out our supplies. We did not even realize this until after discussions with Dr. Johnson and Jakob on why no components were even powering on. We went through a painful process of desoldering the supplies, and we even ended up flipping one of the supplies to the other side of the board because the pads were too damaged on the top side.

After that, the board still did not work. We continually used the multimeter at the output to check whether the voltage was boosted correctly, and we also listened for buzzing from the inductor and power supplies. The voltage remained at V_{in} , and buzzing was observed when the circuit was turned on. We discovered that the MOSFET was also burned out on the high side, and had to replace that. We determined this by performing continuity testing with a multimeter and found that the pins were shorted.

Next, we also had to replace our ADC because some of its pins were shorted after performing additional continuity tests with the multimeter. This became a problem since we now ran the full test at 24 V directly, but the first resolder attempt of the ADC also shorted two pins, causing it to blow immediately and release smoke. We had to resolder this very carefully.

Finally, the circuit worked. We decided to leave the lab and do measurements next week. However, when we returned, once again, no components would turn on, including the gate shutdown switch LEDs and the C2000. We used the test probes to identify which points were an issue. In this case, we discovered that, once again, the 3.3 V pins had a problem.

After probing the test points using a multimeter, we realized that they were only getting 0.7 V. This was strange, and at first we assumed the power supplies had an issue again. However, after discussion with Changxu, we realized that the actual issue was a poor solder joint between the power supply legs and the board. The legs of the power supply were giving 3.3 V, but the board was not receiving it all, likely due to a high-impedance joint. We confirmed this by seeing that a continuity test between the power supply leg itself and the board did not trigger the multimeter. After a couple of attempts at re-flowing and adding more solder, the 3.3 V supply came back online.

After this, we finally turned on the 24 V supply and threw the gate driver switch, and the board worked! We once again used a multimeter to get easy readings of the output voltage with banana jacks. The output returned a solid 47.5 V, and there was no noticeable noise coming from the board. The outputs all looked correct (as can be seen in Figure 10). We used the current probe for I_L and passive probes for the remaining waveforms, using test points.

At the end of this lengthy process, we finally returned to our original result: a working boost converter! We ramped up the load to the full 250 W and encountered no further issues, except for the smell of burning wood from the resistor's waste heat!

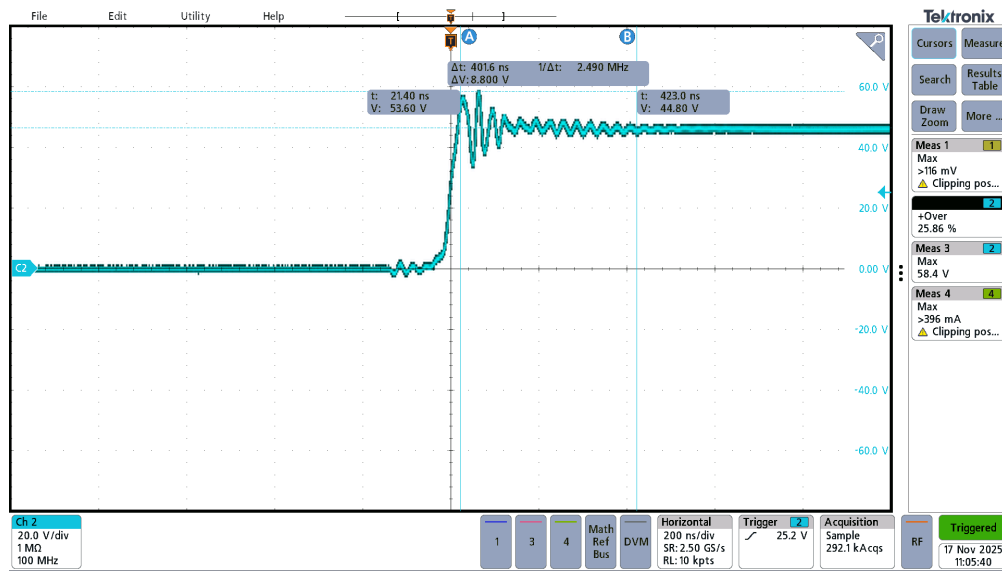


Figure 7: Overshoot on V_{sw}

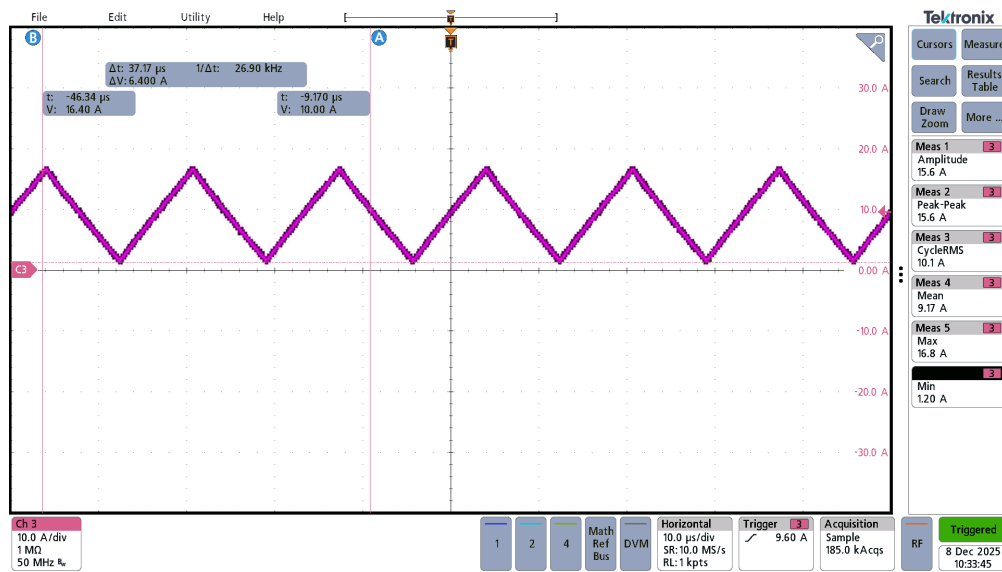


Figure 8: I_L inductor current and ripple

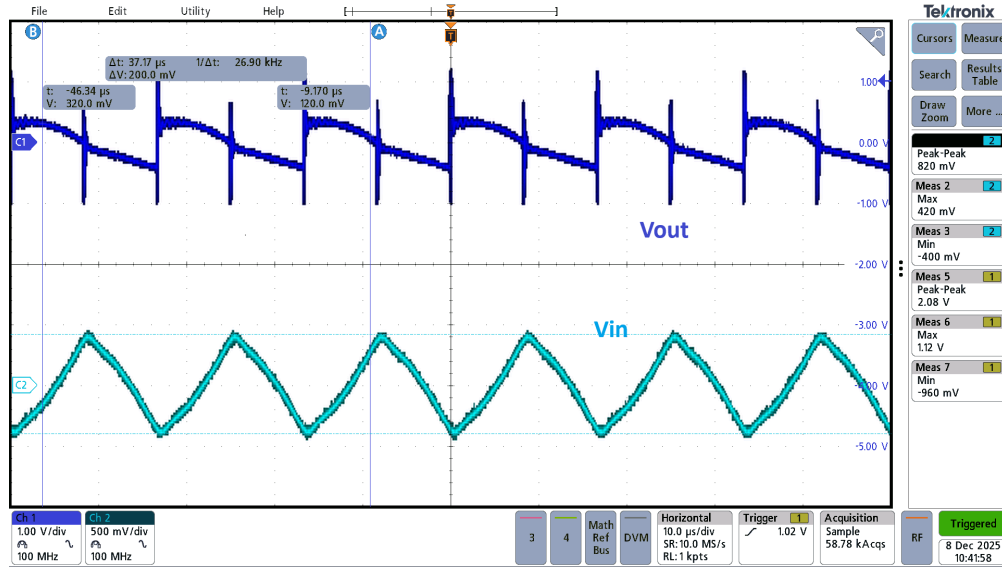


Figure 9: V_o and V_{in} ripple, measured with AC coupling

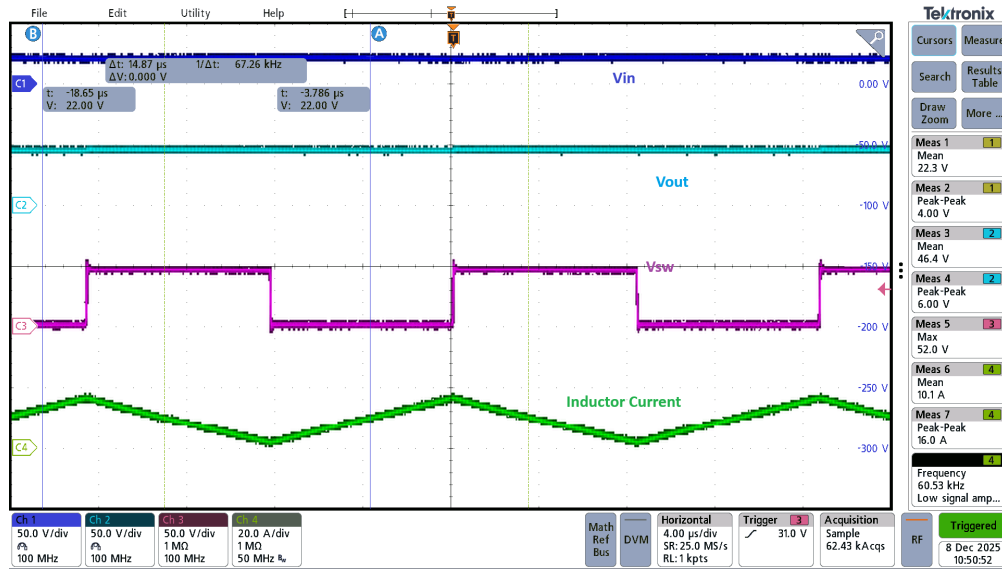


Figure 10: Figure showing V_{sw} , V_o , V_g , and I_L

2.2.2 Load Testing Results Discussion

When it comes to the ripple ratio and inductor current, they line up well with our expected values. The ripple ratio we selected was 80%. The average current was 9.17 A, which is what we also observed on the DC supply, and the peak-to-peak current was 15.6 A, or 7.8 A in one direction, as seen in Figure 8. Calculating by hand, this is a ripple of around 85%, very close to what we had originally! In addition, the ripple is even closer to theoretical, looking at Figure 10, where a mean of 10.1 A and a peak-to-peak of 16 A leads to a ripple ratio of 79%!

The rest of the waveforms look decent. The input voltage was slightly below 24 V, so the output was as well, but they were overall very close to nominal, and the switching waveform looked relatively clean. The output and input voltages also had ripples of only around 2.08 V and 0.820 V, respectively!

2.3 ADC

The ADC was the last section we tackled. The code used to convert the ADC inputs to outputs is given below.

Listing 1: ADC Voltage Calculation

```
1 Vout = (R1+R2)/R2 * AdcaResultRegs.ADCRESULT0 * Vfs /bit_range;  
2 Vin = (R5 + R6)/R6 * AdcaResultRegs.ADCRESULT1 * Vfs /bit_range;  
3 iL = 1/G * ((AdcaResultRegs.ADCRESULT2) / bit_range * Vfs - Vq);
```

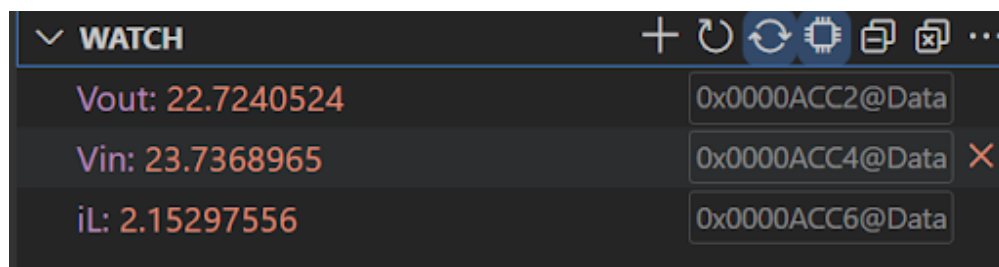


Figure 11: Screenshot of ADC Output when PCB is OFF

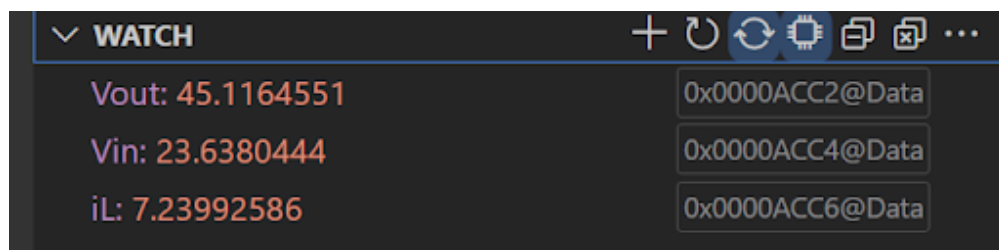


Figure 12: Screenshot of ADC Output when PCB is ON

Our successful ADC implementation is shown above in Figures 11 and 12. We noted that the ADC I_L output was zero when the circuit was disconnected entirely, save for the C2000.

When the gate driver switch was off, but the rest of the circuit was powered, I_L closely matched the DC power supply reading. However, when the PCB was running at full load, our I_L from the DC power supply read 9.68 A, meanwhile our measured ADC was only 7.24 A. After some double-checking and discussion of the code with Jakob, we concluded that this is likely due to ADC variance and that to get a more accurate result, we would likely need some ADC linearization and calibration. Since it was deemed acceptable, we went ahead with this result.

Meanwhile, the voltages line up very well with the multimeter measurements, if a bit lower. This is likely due to the resistors' tolerances in the resistor divider and other unaccounted-for resistances (e.g., in the traces), as well as possible EMI/EMC noise.

2.4 Efficiency Testing

Our peak and full-load efficiencies were the same at 94.67%.

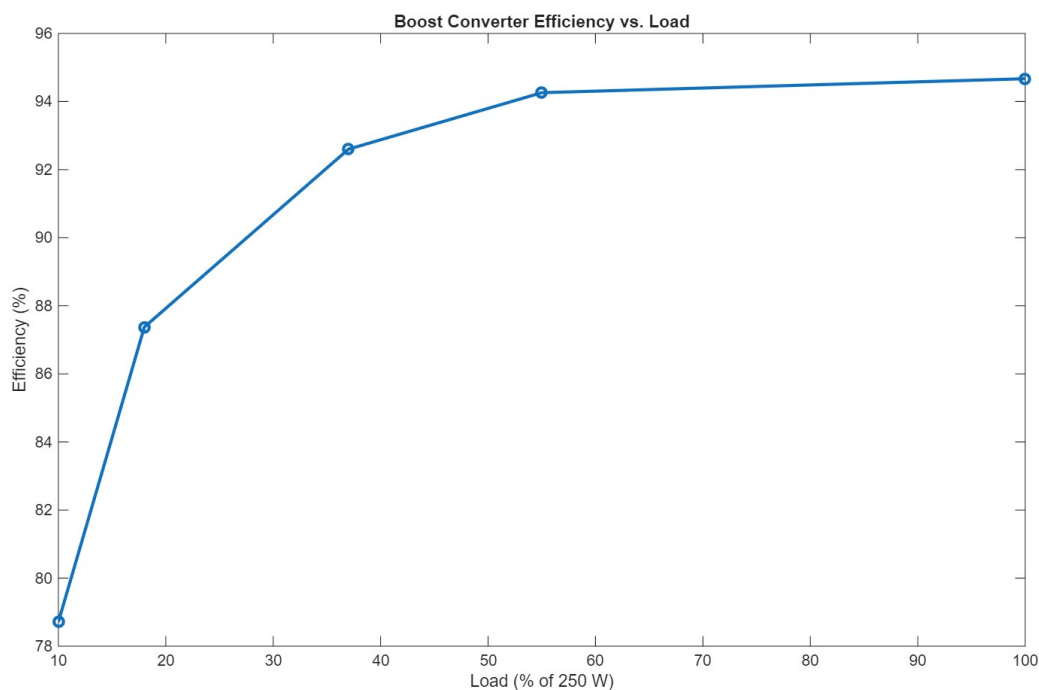


Figure 13: Efficiency of Boost Converter

3 Conclusion

3.1 What did we learn?

We have definitely learned a lot about a boost converter, from how it works conceptually in circuit drawings to how to bring that concept to a tangible product in real life through multiple design, assembly, and testing phases.

More specifically, during the design phases, we learned how to use KiCAD to strategically lay out the PCB based on the given boost converter circuit design. We have also learned how to optimize our boost converter design based on our specific goal and used the optimized ripple ratio and switching frequency values to select the MOSFET device and to calculate four different inductor parameters, such as number of turns, winding resistance, size factor, and air gap for a suitable inductor design.

Regarding the assembly phases, we learned the most efficient way to solder SMDs: use both solder paste and a heat gun to solidify the connection. We have also learned more about programming and embedded system applications in Power Electronics. By controlling the register's value, we can manipulate the ePWM waveform generated from the C2000 microcontroller to control the MOSFET gates as well as perform ADC sampling to read output voltage V_{out} , input voltage V_g , and inductor current I_L .

During the testing phases, we learned and practiced using various lab equipment, including a power supply, an oscilloscope, and a power analyzer, along with different types of probes, to measure the boost converter's performance metrics. We also practiced troubleshooting skills and applied an engineering mindset to identify the fault causing the boost converter not to work correctly, and then used a DMM to trace the issue down slowly, and then isolated it to fix it.

3.2 What did we enjoy?

We enjoyed the project's comprehensive nature, which required ownership of the entire product development cycle. Unlike simulation-only projects, the hands-on process of soldering the board and testing different converters' metrics at physical connections gives us a deeper appreciation for Design for Manufacturability and more intuition and experience for improving the converter next time in terms of weight and efficiency.

Additionally, gaining proficiency with the industry-standard microcontroller and firmware programming to configure it to generate precise ePWM waveforms. We found it rewarding to move beyond analog control and implement digital control techniques through embedded systems.

3.3 What was the most difficult part?

The primary fabrication bottleneck is definitely soldering components onto the PCB and desoldering them during replacement, due to tight component spacing and the intricate process of removing excess solder from component holes.

3.4 What did you dislike about the project? What do you think can be done better, either administratively or just in your design? Provide suggestions for your future designs or future iterations of this course.

This project provides us with an excellent opportunity to practice engineering skills that will be valuable to the industry. The project can be handled more effectively administratively by carefully reviewing the project guidelines handout and making necessary adjustments before sending it to the students. Doing this will prevent misunderstandings among students that could lead to significant errors, requiring the board to be rebuilt from the start.

3.5 What would you have done differently if you knew everything you know now?

The first thing is not to slowly ramp up the voltage and burn out the power supplies. That error alone cost us essentially all of our debugging time. We went from a working board to two weeks of debugging to get back to where we began.

Other than that, there are many things we would have done differently. One thing we found out the hard way was that putting test points in bad places makes them very hard to access. This was especially true near the critical loop, and it also became a problem when we placed a few under the C2000.

Another improvement would have been to be more careful of solder joints from the beginning. We struggled with some improper connections, mainly due to insufficient heat.

The most significant change would be to be more ambitious with the board. We played it safe by using a large PCB because our team was inexperienced with soldering, assembly, and debugging. We wanted to ensure the form functioned properly. However, now that we have more experience, we aim for a much more compact board next time.

3.6 Any particular highlights?

The highlight was demonstrating a fully functional prototype despite significant fabrication and rework challenges by successfully powering the E-bike at a stable, regulated 48V in the real world. Seeing the Ebike operate flawlessly is the ultimate validation of our design, proving that our converter can handle real-world demands.